

ENBC321/BIOE658I: Homework #3

Due: Monday, 02/23/26 by 5:00 PM, beginning of the class

Submit your homework to ELMS, including Python code and the dataset

Overall points: 100

Note: Late homework may be submitted up to 24 hours after the due date, but a 50% penalty will be applied.

Note: Please aim to write clean, well-organized code that is easy for others to understand. Be sure to include comments or explanations where needed. Points may be deducted if this is not done.

Note: Please name your codes using the following format: `problem1_fullname.ipynb`. Start with the problem number and followed by your full name, e.g., **`problem1_jimmyazarnoosh`**.

Note: Using generative AI for learning purposes is allowed. However, directly copying or closely replicating AI-generated content is strictly prohibited and will result in a grade of ZERO. Repeated violations may lead to more severe consequences.

Note: Submit **a single zip file** containing the following:

- The dataset you used
- Python code files in ***.ipynb**
- Your answers to the questions in PDF format

Feature Selection Techniques

In this assignment, you will perform feature selection on an online dataset (from Kaggle or any other website) containing at least five but no more than ten features.

Problem 1. Your goal is to evaluate different feature selection methods (listed below) and **determine the best approach** for selecting important features. Rank these methods from best to worst and justify your ranking. (70 points)

For each feature selection method:

- Identify and report the top four most important features.
- Report the R-squared (R^2) and Mean Squared Error (MSE) for each method.

Feature Selection Methods:

1. All-in (or “Full Model”) Selection (10 points)
2. Backward Elimination (10 points)
3. Forward Selection (10 points)
4. Bidirectional Elimination (10 points)
5. ANOVA F-test (10 points)
6. Mutual Information (10 points)
7. Pearson Correlation (10 points)

Prediction Using Support Vector Regression

Problem 2. You are given a dataset ([gene expression data.csv](#)) containing experimental values of gene expression and the corresponding protein expression levels. Your task is to apply Support Vector Regression (SVR) to **predict protein expression levels** based on gene expression. (30 points)

The dataset contains the following columns:

Gene_Expression (X): Expression levels of a gene

Protein_Expression (y): Measured protein expression levels

Instructions:

- Load the dataset using **pandas**. The dataset is in a CSV file, so use `pd.read_csv()`.
- **Split the data** into training and testing sets (80% training, 20% testing).
- **Fit an SVR model** from the `sklearn.svm` module to the data.
- **Train the SVR model** on the training data.
- **Predict protein expression levels** on the test set.
- **Plot:**
 - The true protein expression values vs. the predicted values for the test set.
 - The SVR regression curve along with the data points.

Tasks:

- Compute the Mean Squared Error (MSE) and the R^2 score for the model. (5 points)
- Use both **linear** and **nonlinear** kernels and visualize each in a separate plot. (5 points)
- For both kernels, experiment with different epsilon values: $\epsilon = 0.1, 2, 4, 6, 8, 10$. (10 points)
- Analyze/summarize how different ϵ values affect regression fit and prediction accuracy in both kernel types. (5 points)
- Plot the results according to the instructions. (5 points)